

Lecture 5:

Data Collection 4: Social Media Data Sampling + Projects 101

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

Recap: APIs and Platform Affordances

APIs

- Application Programming Interface
- Typically not designed for research purposes (therefore not tailored to research needs)
- Several platforms offer APIs as access points
- Often you have to register, sometimes also accept terms of services

APIs are typically not designed for research

- Core Purpose of Social Media APIs
 - Allow developers to interact with platform features
 - Expand the platform's ecosystem beyond its own website or app, e.g., [Reddit Mod tools](#),
 - Social login (“Sign in with Facebook”) and sharing content directly from other apps
 - Social media management tools (e.g., schedulers)
 - ...
- To use an **official** social media API, you must:
 - Create an account
 - Register as a developer
 - Create an app
 - Authenticate users
 - Obtain access tokens — **And be careful not to share them with anyone!**
 - Respect permissions & rate limits

APIs vs Web Scraping

Category	Web Scraping	API-Based Data Collection
Access Method	Extracts data from website front-end (HTML pages)	Sends structured requests to official backend endpoints
Official Support	Not officially supported by platforms	Officially provided by platforms like Facebook, Instagram, X, and LinkedIn* Unofficial APIs also exist but much less stable
Data Format	Unstructured HTML → requires parsing	Structured JSON/XML
Stability	Breaks when website layout changes	Stable and version-controlled
Authentication	May bypass login systems	Requires secure login (OAuth, tokens) [official APIs!]
Permissions	No formal permission model	Uses scoped access permissions
Rate Limits	No official limits, but risk of blocking	Clearly defined request limits
Legal Risk	Potential Terms of Service violations	Governed by developer agreements
Maintenance Effort	High – frequent updates required	Lower – documented and supported
Ethical Considerations	May raise privacy concerns	Designed with privacy safeguards

Platform Affordances

- Action possibilities enabled by an environment [Gibson, “[The Senses Considered as Perceptual Systems](#)”]
- They shape how people create, share, and interact
 - with content
 - with other users
- Influence visibility and identity
- Related concepts:
 - Platform design, features*, governance, display

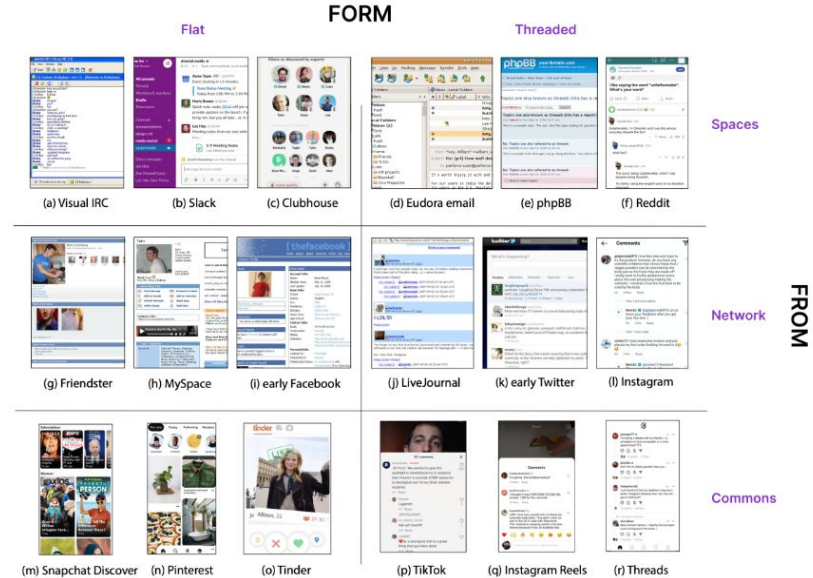


Fig. 4. Examples of systems both old and current in each cell of the Form-From model.

“[Form-From: A Design Space of Social Media Systems](#)” Zhang et al., 2024

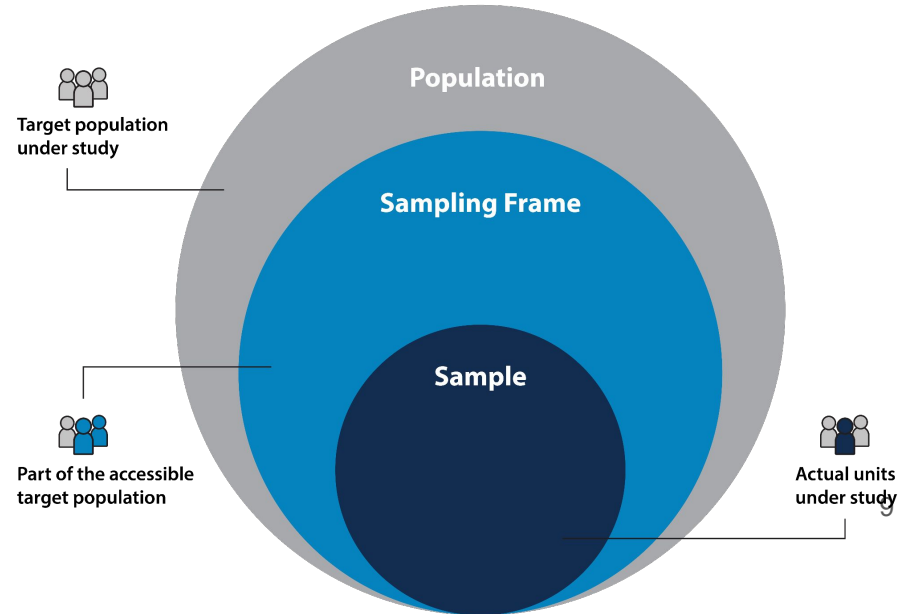
What do platform affordances imply for SMDA?

- Platform affect user behavior and the data they generate on these platforms
- Have to be especially careful during sudden affordance changes
 - Are your findings of platform behavior affected by changes?
- On the flipside: changes in the platform are excellent opportunities to study the impact of these affordances
 - Can also be done retroactively
 - See studies in the “resources” section of today’s lecture slides

Today: **Sampling Social Media Data**
+ Projects 101

Sampling in traditional social research

- **Sample:** A subset of instances (e.g., posts, users) selected from a target population to represent it in a study
- **Sampling frame:** The list or set of all instances from which the sample is actually drawn (e.g. a voter registry, a customer database) — ideally matches the target population
- How large should the sample be?
 - Depends on desired statistical power, effect size, and acceptable margin of error
 - Larger samples reduce sampling error, but larger samples != better samples
- When is a sample considered good?
 - When it is **representative** of the population: low bias (systematic error) and low variance (random error) — ideally achieved through random sampling



<https://www.nlm.nih.gov/oet/ed/stats/02-100.html>

Do we have to sample social media data?

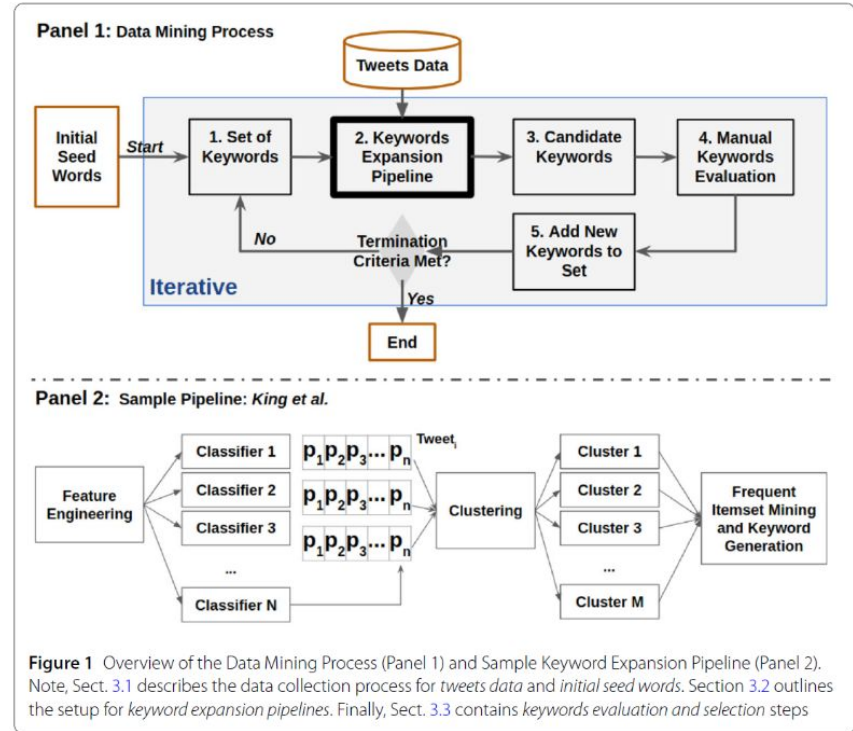
- Often yes; API access is usually rate-limited, meaning you only retrieve a fraction of all posts
- Cost, storage & compute limits: even when full data is obtainable, processing billions of posts requires infrastructure most researchers don't have
- Platform-imposed sampling: some APIs return a random sample by default (e.g. Twitter's old 1% "Spritzer" stream), so sampling happens whether you choose it or not
- the sample you get is not always a sample you designed, raising questions about representativeness and bias

What are different ways in sampling social media data?

- Depends on what you're trying to study and the platform features and affordances → determines the sampling frame
- For studies of the entire platform — the entire platform is the sampling frame
 - Get the data directly from the platform
 - Do large-scale web crawling similar to the Truth social paper, but can still miss some people
 - Use external lists of information to create the sampling frame [[Wagner et al., 2015](#)]
- Can be keywords or hashtags, especially for specific topics, e.g., protests
- Can be based on specific communities, e.g., advice-related subreddits
- Can be a list of accounts
- ...

Discuss: Can sampling affect data quality?

- What are the impacts of different sampling decisions?
- If we want to collect data about a specific event, e.g., a protest, what is the best data collection strategy?



Bozarth, Lia, and Ceren Budak. "[Keyword expansion techniques for mining social movement data on social media](#)." EPJ Data Science 11.1 (2022): 30.

Today: Sampling Social Media Data **+ Projects 101**

Reminder: Project Score distribution

1. Project pitches [5%] → 5 mins per teams
2. Midway Presentation [15%] → 15 mins per team
3. Final reports [60%]
 - a. All reports should include: introduction, related work, experiments, results, conclusion
 - b. Team member contributions: all members of a group need not get the same grades
 - c. Include a link to a github repository with your code and dataset

What do people do in a research project?

- Try to study something that piques their curiosity
 - What causes conflict in homogenous online communities?
 - How do people express their identity in anonymous platforms?
- Ideally, this should not only satisfy their own curiosity but also have societal impact

What do people do in a research project?

- Try to study something that piques their curiosity
 - What causes conflict in homogenous online communities?
 - How do people express their identity in anonymous platforms?
- Ideally, this should not only satisfy their own curiosity but also have societal impact
- For quantitative social research specifically (order may differ):
 - Start with a research question, e.g., “**do protest related tweets have more informal language?**”
 - Describe concepts and constructs related to the research questions
 - Describe the operationalization of concepts and constructs, i.e., how it will be measured
 - e.g., variable p signifies the volume of protest-related tweets on a given day
 - We measure $p = \#$ tweets with at least two protest-related keywords
 - Collect data that lets you study your research question
 - Statistically and computationally analyze the data, e.g., measure formality in *both* protest and non-protest tweets
 - Create summary statistics and visualizations to demonstrate your answer to the research question

What do people do in a research project?

- Confirmatory vs. exploratory approach

- Exploratory:

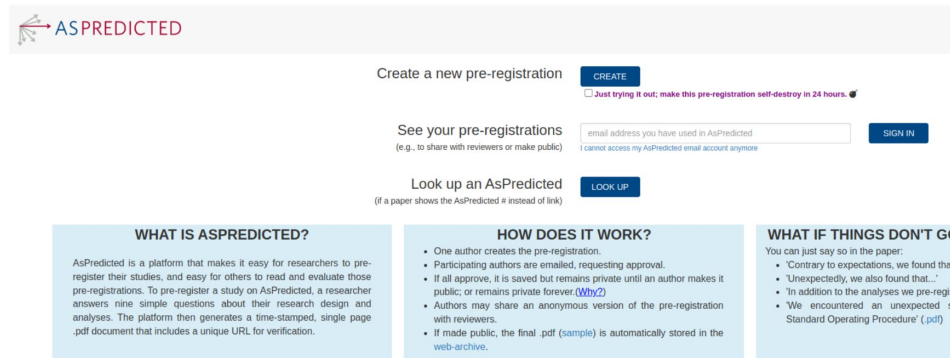
- Start with a broad research question
- Novel data source
- Data-driven study of the novel data

- Confirmatory:

- Start with a concrete research question and theory
- Confirm whether a new setting conforms to this theory
- Good to preregister

- In practice: often a mix of the two

- For an exploratory project: see if exploratory findings adhere to existing theories
- For a confirmatory project: try and explore why data does not adhere to existing theory



The screenshot shows the AsPredicted website interface. At the top, there is a logo with a red arrow and the text "ASPREDICTED". Below the logo, there are three main sections:

- Create a new pre-registration**: This section has a "CREATE" button and a link that says "Just trying it out; make this pre-registration self-destroy in 24 hours."
- See your pre-registrations**: This section has a text input field for an email address and a "SIGN IN" button. Below the input field, there is a note: "email address you have used in AsPredicted (e.g., to share with reviewers or make public) cannot access my AsPredicted email account anymore".
- Look up an AsPredicted**: This section has a "LOOK UP" button and a note: "(If a paper shows the AsPredicted # instead of link)".

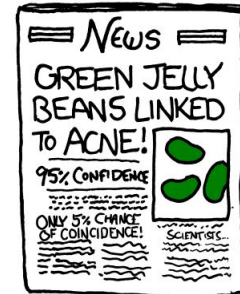
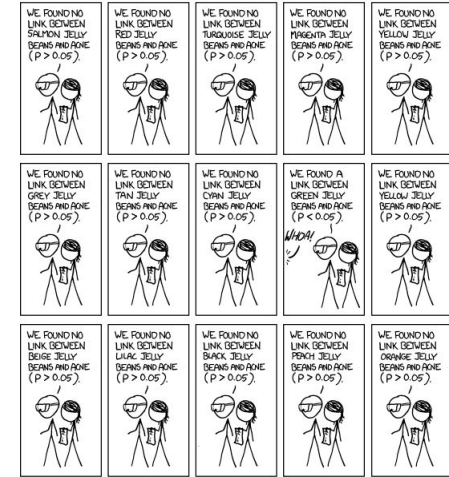
At the bottom, there are three informational boxes:

- WHAT IS ASPREDICTED?**: This box explains that AsPredicted is a platform that makes it easy for researchers to pre-register their studies and easy for others to read and evaluate those pre-registrations. It also mentions that the platform generates a time-stamped, single page .pdf document that includes a unique URL for verification.
- HOW DOES IT WORK?**: This box lists four steps: 1. One author creates the pre-registration. 2. Participating authors are emailed, requesting approval. 3. If all approve, it is saved but remains private until an author makes it public; or remains private forever. 4. Authors may share an anonymous version of the pre-registration with reviewers. It also mentions that if made public, the final .pdf (sample) is automatically stored in the web-archive.
- WHAT IF THINGS DON'T GO AS EXPECTED?**: This box explains that you can just say so in the paper. It lists three examples: 1. 'Contrary to expectations, we found that...'. 2. 'Unexpectedly, we also found that...'. 3. 'In addition to the analyses we pre-registered, we also found that...'. It also mentions that if encountered an unexpected outcome, it should be documented in the Standard Operating Procedure (.pdf).

<https://aspredicted.org/>

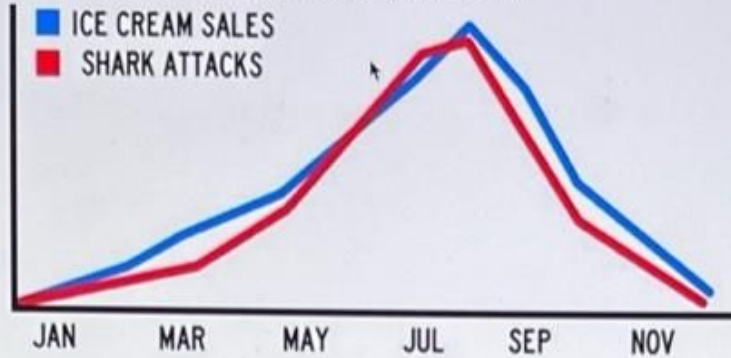
What **should** people **not** do in a research project?

- Collect data and analyze it in a manner that conforms to their own biases or specific outcome
- p-Hacking: trying many statistical tests and only reporting those that are significant
- HARKing: Hypothesizing After the Results are Known
 - Changing the hypothesis based on what you find without stating the original hypothesis



<https://xkcd.com/882/>

This graph shows that as ice cream sales increase shark attacks also increase. We all know that ice cream isn't good for you, but I had no idea that ice cream attracted sharks!
Which of these is the correct analysis of this graph?



- ☐ Ice cream should be banned from the beach as it causes shark attacks.
- ☐ There is a clear correlation between ice cream and shark attacks. Ice cream is protective against shark attacks so all beachgoers should be encouraged to eat ice cream.
- ☐ People who suffer shark attacks in the summer reduce stress by eating ice cream.
- ☐ Ice cream intake and ocean beach-going happen most often during sunny weather. They are correlated but one does not cause the other.

<https://www.chegg.com/homework-help/questions-and-answers/graph-shows-ice-cream-sales-increase-shark-attacks-also-increase-know-ice-cream-isn-t-good-q85332558>

What **should** people do in a research project?

- Control for confounding variables (more on this when we get to data analysis)
 - Hidden variables that might explain the relationship you are studying but you haven't considered them, e.g., warm weather impacts both shark attacks and ice cream sales
- Preregistration is highly recommended for confirmatory analysis setups
 - Also encouraged for exploratory setups
 - Your project pitch serves as preregistration of your project. You can deviate from it (especially based on instructor feedback) but you have to justify why
- **Reflexivity:** we cannot account for all biases (in ourselves, in data, in analysis) but we can try to reflect on them and document them
- Reproducibility materials: ensure that others can reproduce your work by giving them access to your data and code

What type of project are you expected to do in this course?

- Describe a clear research question or hypothesis, e.g., “Is there more anti-AI content on Bluesky than pro-AI content?”
 - Break it down into related subparts, e.g.,
 - RQ1.1: “Has anti-AI content on Bluesky increased over time? ”
 - RQ1.2: “What are common themes in anti-AI posts on Bluesky?”
 - RQ1.3: “Are posts more anti-AI or replies?”
- Collect data from the platform
 - Programmatically e.g., collect a representative sample of AI-related Bluesky conversations using keywords like ‘AI’, ‘LLMs’, etc
 - Make sure your dataset is able to answer your research question

What type of project are you expected to do in this course?

- Analyze the data computationally and at scale
 - E.g., automatic stance detection that labels whether a post is pro or anti-AI
- Aggregate your results and conduct descriptive analysis
 - E.g., over time, in different communities, etc
 - Make sure you account for confounders, e.g., pro-AI advocates using different terms
 - Make sure to use the right statistical measures depending on the nature of your data
- Validate your data analysis methods
 - E.g., by manually annotating a small sample and establishing the performance of your computational methods
 - make sure the method works equally well for pro-AI or anti-AI posts
- Discuss the implications of your results, e.g., the common reasons people are against AI on Bluesky

Let's see some examples

- **Research Question:** Is there gender bias in Wikipedia articles about women?
- **Construct:** gender bias
- **Platform:** Wikipedia
- **Data Collection Strategy:** start with an external list of notable people and then match with Wikipedia
- **Data Analysis Strategy:**
 - statistical analysis of coverage
 - lexical analysis of how women's and men's articles differ
 - network analysis of articles to see links to articles
- **Results:** women and men have similar coverage, but women are described differently (in relation to other men)

It's a Man's Wikipedia? Assessing Gender Inequality in an Online Encyclopedia

Claudia Wagner
GESIS & Uni. Koblenz
claudia.wagner@gesis.org

David Garcia
ETH Zürich
dgarcia@ethz.ch

Mohsen Jadidi
GESIS
mohsen.jadidi@gesis.org

Markus Strohmaier
GESIS & Uni. Koblenz
markus.strohmaier@gesis.org

Abstract

Wikipedia is a community-created encyclopedia that contains information about notable people from different countries, epochs and disciplines and aims to document the world's knowledge from a neutral point of view. However, the narrow diversity of the Wikipedia editor community has the potential to introduce systemic biases such as gender biases into the content of Wikipedia. In this paper we aim to tackle a sub problem of this larger challenge by presenting and applying a computational method for *assessing gender bias on Wikipedia along multiple dimensions*. We find that while women on Wikipedia are covered and featured well in many Wikipedia language editions, *the way women are portrayed* starkly differs from the way men are portrayed. We hope our work contributes to increasing awareness about gender biases online, and in particular to raising attention to the different levels in which gender biases can manifest themselves on the web.

Introduction

Wikipedia aims to provide a platform to freely share the sum of all human knowledge. It represents an influential source of information on the web, containing encyclopedic information about notable people from different countries, epochs and disciplines that is used for learning and educational purposes worldwide. Wikipedia is also a community-created effort driven by a self-selected set of editors. The demographic characteristics of this set of editors is known: it is predominately white and male (Lam et al. 2011; Collier and Bear 2012; Hill and Shaw 2013).

This known gender bias in the population of editors has the potential to introduce gender biases into the contents of Wikipedia as well. For example, the population bias might lead to differences in the ways women and men are por-

of gender biases on Wikipedia. Yet, due to the influential nature of Wikipedia, it is important to reveal, assess and correct such biases, if they exist. This paper tackles a sub-part of this larger challenge.

Objectives: In particular, the overall goal of this work is to *assess potential gender inequalities in Wikipedia articles along different dimensions*.

Approach: To assess the extent to which Wikipedia suffers from potential gender bias, we analyze articles about notable people in six language editions along four different gender bias dimensions: coverage bias, structural bias, lexical bias and visibility bias. *Coverage bias* determines differences between the number of notable women and men portrayed on Wikipedia. For example, one might hypothesize that notable men are more likely to be covered by Wikipedia. *Structural bias* quantifies gender homophily/disassortativity, i.e. gender-specific tendencies to preferably link articles of notable people with the same or different gender. For example, one might hypothesize that articles about women have more links to men than vice versa. *Lexical bias* reveals inequalities in the words used to describe notable men and women on Wikipedia. For example, articles about women are potentially more likely to mention their family (husband or kids) than articles about men. *Visibility bias* reflects how many articles about men or women make it to the front page of Wikipedia. Again, one can hypothesize that articles about men might have better chances to be selected.

Contributions & Findings: We present and apply a computational method for *assessing gender bias on Wikipedia along multiple dimensions*. We find that most Wikipedia language editions exhibit a slight over-representation of women, but the representational differences in the coverage of women are not significant. That means, men and women are featured equally in all six Wikipedia language editions. However, we do not find significant differences in the selection procedure of the English Wikipedia editor community is sensible to gender in-

Wagner, Claudia, et al. "It's a man's Wikipedia? Assessing gender inequality in an online encyclopedia." *Proceedings of the international AAAI conference on web and social media* Vol. 9. No. 1. 2015. (www.aaai.org). All rights reserved.

Let's see some examples

- **Research Question:** Are comments on right-leaning Youtube videos moderated more?
- **Construct:** political bias in moderation
- **Platform:** Youtube
- **Data Collection Strategy:** Videos which had been fact-checked by Snopes and PolitiFact
- **Data Analysis Strategy:** statistical and linguistic comparison of
 - right vs. left-leaning videos
 - misinformative vs. non-misinformative videos
- **Results:** differences in moderation between right and left-leaning videos goes away when you control for confounding variables like hate speech in comments

Bias Misperceived: The Role of Partisanship and Misinformation in YouTube Comment Moderation

Shan Jiang, Ronald E. Robertson, Christo Wilson

Northeastern University, USA
{sjiang, rer, cbw}@ccs.neu.edu,

Abstract

Social media platforms have been the subject of controversy and scrutiny due to the spread of hateful content. To address this problem, the platforms implement content moderation using a mix of human and algorithmic processes. However, content moderation itself has led to further accusations against the platforms of political bias. In this study, we investigate how channel partisanship and video misinformation affect the likelihood of comment moderation on YouTube. Using a dataset of 84,068 comments on 258 videos, we find that although comments on right-leaning videos are more heavily moderated from a correlational perspective, we find no evidence to support claims of political bias when using a causal model that controls for common confounders (e.g., hate speech). Additionally, we find that comments are more likely to be moderated if the video channel is ideologically extreme, if the video content is false, and if the comments were posted after a fact-check.

1 Introduction

In the wake of 2016–2017 global election cycle, social media platforms have been subject to heightened levels of controversy and scrutiny. Besides well-documented privacy issues (Solon 2018), social media platforms have been increasingly used to promote partisanship (Allcott and Gentzkow 2017), spread misinformation (Constone 2018), and breed violent hate speech (Ofteanu et al. 2018).

The solution promulgated by social media platforms for this is an increase in content moderation. In terms of

censoring their views (Kamisar 2018; Usher 2018). Two US House Committees have held hearings on content moderation practices to “specifically look at concerns regarding a lack of transparency and potential bias in the filtering practices of social media companies (Facebook, Twitter and YouTube)” (Bickert, Downs, and Pickles 2018; Dorsey 2018). These concerns are driven by multiple factors, including anecdotal reports that Facebook’s Trending News team did not promote stories from conservative media outlets (Nunez 2016), Twitter “shadow banned” conservative users (Newton 2018), fact-checking organizations are biased (Richardson 2018), and selective reporting by partisan news agencies (Arceneaux, Johnson, and Murphy 2012).

However, there is, to our knowledge, no scientific evidence that social media platforms’ content moderation practices exhibit systematic partisan bias. On the contrary, there are many cases where ideologically liberal users were moderated, although these cases have received less attention in the media (Masnick 2018). It is possible that moderation only appears to be biased because political valence is correlated with other factors that trigger moderation, such as bullying, calls to violence, or hate speech (Gillespie 2018). Further, there is evidence suggesting that users tend to overestimate bias in moderation decisions (Shen et al. 2018).

In this study, we use YouTube as a lens and take a first step towards disentangling these issues by investigating how partisanship and misinformation in videos affect the likelihood of comment moderation. Specifically, we examine four hypotheses related to four variables of YouTube videos:

Jiang, Shan, Ronald E. Robertson, and Christo Wilson. “[Bias^{left/right} misperceived: The role of partisanship and misinformation in youtube comment moderation](#).” *Proceedings of the International AAA/checked (11/20) Conference on Web and Social Media*. Vol. 13. 2019. ^{with the null hy-} use correlational and causal models to collect evidence on rejecting the null hypotheses. The conceptual framework of our hypotheses is shown in Figure 1.

2018; Facebook 2018; Twitter 2018).

This increased reliance on content moderation faces a

How do you go about doing the project?

1. The ideal pipeline: Research question → platform selection → data collection → data analysis → profit!
2. However, sometimes the original plan might not work because you don't get the data you expected
3. Start with a research question and think about what data you want to collect to answer that question
4. You can reuse existing datasets that other researchers have created
 - a. Check <https://dataverse.harvard.edu/> or <https://www.kaggle.com/datasets>
 - b. But you get bonus points for collecting a dataset of your own

How do you go about doing the project?

1. The ideal pipeline: Research question → platform selection → data collection → data analysis → profit!
2. However, sometimes the original plan might not work because you don't get the data you expected
3. Start with a research question and think about what data you want to collect to answer that question
4. You can reuse existing datasets that other researchers have created
 - a. Check <https://dataverse.harvard.edu/> or <https://www.kaggle.com/datasets>
 - b. But you get bonus points for collecting a dataset of your own
5. **Important:** It will be difficult to do a computational project using your laptops without access to servers. Check the first exercise and apply for access to the BWUniCluster ASAP!

On what basis will your project be graded?

1. how 'good' your results are
 - good != positive direction, effect size, statistical significance, etc
 - but how robust, replicable, and reliable your results are
 - negative results are also results
 - how well your analysis supports your claims
2. research design [ensuring data and measurement quality, controlling for confounders, validation]
3. theoretical, statistical, and practical rigor

All project and grading information can be found in the [course github repo](#):

1. Grading rubric
2. Project ideas
3. AI usage policy

Summary and Next Week

- This session we learned:
 - Sampling in traditional social research
 - How to sample social media data
 - How sampling impacts data quality
 - Project Information
- Next time:
 - Data Cleaning
 - Wikipedia data collection and cleaning
 - Readings:
 - 1. Sen, Indira, ["Social Media Data Cleaning"](#)
 - 2. Bakhshi, Saeideh, David A. Shamma, and Eric Gilbert. ["Faces engage us: Photos with faces attract more likes and comments on instagram."](#) Proceedings of the SIGCHI conference on human factors in computing systems. 2014.

Questions?